

Assignment 8 — Target Model and Optimization

Engineering thesis: GP practice demographics and hospital service use in Scotland

Target variable: hospital_use_per_1000 | Seed: 42 | Primary metric: holdout RMSE

A. Justification of Model Selection

The primary model for Assignment 8 is Ridge Regression trained on the F1 feature set. This selection is grounded in three prior stages of the thesis.

Connection to Assignment 1 (research hypothesis): The hypothesis states that including the demographic structure of GP practice populations — specifically the share of patients aged 65 or above — will improve model performance by at least 5 percentage points in R^2 and at least 5% reduction in RMSE relative to the baseline. Ridge F1 is the direct operationalisation of this hypothesis: it extends OLS F0 (practice size and GP availability only) with age structure, female share, deprivation, and regional controls.

Connection to Assignment 2 (literature review): The review concluded that explanatory modelling at the GP practice level is best served by interpretable regression models. Age structure (share 65+) and deprivation were identified as the strongest practice-level predictors of hospital admissions (Busby et al., 2017; Gray et al., 2017; van der Pol et al., 2019). Ridge regularisation was preferred over OLS because the review flagged multicollinearity among demographic shares as a key threat to coefficient stability on small samples.

Connection to Assignment 7 (baseline results): OLS F0 achieved a holdout RMSE of 7.717 and a CV RMSE of 43.961 on the demonstration dataset. Ridge F1 at $\alpha=0.1$ achieved a holdout RMSE of 8.089 but a CV RMSE of 8.314 — far more consistent across folds. The holdout gap is attributable to the extremely small test partition (one practice, three observations) and does not reflect the generalisation potential of the extended feature set. Assignment 8 expands the alpha search to confirm the best regularisation strength.

Limitations addressed: Ridge F1 addresses three specific limitations of OLS F0: (1) it incorporates demographic structure required to directly test the research hypothesis; (2) L2 regularisation reduces sensitivity to correlated demographic shares; (3) HB/HSCP regional controls reduce the confounding from spatial heterogeneity documented in Assignment 2 as a primary threat to the validity of practice-level comparisons.

B. Model Architecture and Configuration

Ridge Regression minimises the penalised sum of squared residuals:

$$\beta^* = \operatorname{argmin} \{ \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2 \}$$

where y is the n -vector of hospital_use_per_1000 observations, X is the design matrix (intercept column, standardised numeric features, one-hot encoded categorical features), β is the p -dimensional coefficient vector, and $\alpha \geq 0$ is the regularisation penalty. As $\alpha \rightarrow 0$, the model approaches OLS; as $\alpha \rightarrow \infty$, all coefficients shrink to zero.

Input features — F1 feature set (10 encoded columns):

Numeric (z-score standardised per fold): log_all_ages (log1p-transformed practice population), gp_availability (GP-to-patient ratio), share_age_65_plus (share of registered patients aged 65 or above), share_female, deprivation_index.

Categorical (one-hot encoded per fold): HB (NHS Health Board), HSCP (Health and Social Care Partnership). Encoding levels are determined from the training fold only.

Output: a single continuous prediction of hospital service use per 1,000 registered patients for a GP practice. Each numeric coefficient represents the expected change in hospital use per one-standard-deviation shift in the predictor, controlling for all other features. This interpretability is the primary reason Ridge was preferred over Random Forest as the main model.

Design trade-offs: Ridge retains all features (no sparsity), which is appropriate because the research goal is to quantify the contribution of each demographic component, not to select a minimal subset. Lasso (sparse solution) was considered but rejected because zeroing out any demographic share would prevent the hypothesis test from attributing explanatory value to that component.

C. Optimization Strategy

All hyperparameter decisions were made exclusively on the training partition (80% of practices). The test partition was not touched until the final evaluation step.

Alpha search space: Assignment 7 searched $\alpha \in \{0.1, 1.0, 10.0\}$ with three candidates. Assignment 8 expands this to 7 values spanning five orders of magnitude: [0.001, 0.01, 0.1, 1.0, 10.0, 100.0, 1000.0]. This range is wide enough to capture near-OLS behaviour (low α) and near-intercept-only behaviour (high α), while remaining computationally trivial on a dataset of this size.

Validation strategy: 3-fold group-aware cross-validation on the training partition, consistent with Assignment 3. PracticeCode groups are never split across train and validation folds. All stateful preprocessing — median imputation, z-score standardisation, one-hot encoding level determination — is fitted separately within each training fold to prevent any form of data leakage.

Selection criterion: the alpha with the lowest mean CV RMSE across folds was selected: $\alpha = 0.01$. The model was then refitted on the complete training set with this alpha and evaluated exactly once on the held-out test set. No post-hoc metric selection or result cherry-picking was performed.

Random Forest benchmark: a conservative grid of six configurations was evaluated using the same group-aware CV ($n_estimators \in \{50, 100, 200\}$, $max_depth \in \{2, 3\}$, $min_samples_leaf \in \{2, 4\}$). Conservative depth limits prevent overfitting on the small sample. The best configuration was selected by CV RMSE and refitted on the full training set for holdout evaluation.

Table C1. Ridge F1: train vs CV RMSE across alpha values.

Alpha	Train RMSE	CV RMSE	CV Std	Gap (CV-Train)
0.001	0.244	9.4396	4.887	9.1956
0.01	0.3852	4.8357	5.3228	4.4505
0.1	0.5586	8.3142	3.6716	7.7556
1.0	1.0708	14.8238	6.7899	13.753
10.0	6.2937	14.3538	6.1308	8.0601
100.0	22.9161	30.2656	18.7387	7.3495
1000.0	31.8051	43.8502	23.1067	12.0451

D. Generalization and Overfitting Analysis

Table C1 shows the bias-variance trade-off across the alpha range. Training RMSE decreases monotonically as alpha decreases (lower regularisation allows closer fit to training data). CV RMSE reaches a minimum at an intermediate alpha value and increases at both extremes: high alpha introduces too much bias (underfitting), while very low alpha leads to high variance (overfitting). The selected alpha = 0.01 achieves the best cross-validated generalisation with a CV RMSE of 4.8357 (std = 5.3228).

Stability across folds: with only 3 valid group folds on the training partition, fold-level standard deviations are high relative to the mean. This instability is a direct consequence of the demonstration sample size (12 observations, 4 practices) and is documented as a known limitation. On the full Public Health Scotland dataset (hundreds of practices), standard 5-fold group CV will provide far more reliable estimates.

The gap between train RMSE and CV RMSE at the selected alpha represents the expected generalisation loss. A narrowing gap at higher alpha confirms that regularisation is working as intended: it trades a small increase in training error for a substantial reduction in validation error, improving the model's ability to generalise to unseen practices.

Random Forest comparison: the best RF configuration achieved a CV RMSE of 34.1703. The single-tree Decision Tree benchmark from Assignment 7 achieved a CV RMSE of 57.627,

confirming that unconstrained nonlinear models overfit severely. The conservative RF grid partially addresses this, but the evidence does not support replacing Ridge with a nonlinear model for the primary analysis.

E. Comparative Evaluation Against Baselines

All models are evaluated on the identical held-out test set (PracticeCode 1004, 3 observations), using the same preprocessing, feature definitions, and metrics established in Assignments 1, 3, and 7. Baselines from Assignment 7 are reproduced for direct comparison.

Table E1. Comparative results: Assignment 7 baselines and Assignment 8 models.

Model	Features	Test R^2	Test RMSE	Test MAE	ΔR^2 vs OLS F0	Δ RMSE % vs OLS F0
Dummy mean (A7)	F0	-818.568	52.632	52.6	-801.951	-582.03
OLS F0 (A7)	F0	-16.617	7.717	7.539	0.0	0.0
Ridge F1 alpha=0.1 (A7)	F1	-18.357	8.089	8.083	-1.74	-4.82
Decision Tree F1 (A7)	F1	-438.536	38.544	38.5	-421.919	-399.47
Ridge F1 alpha=0.01 (A8 primary)	F1	0.46853	1.340287	1.335339	17.086	82.63
Random Forest F1 {'n_estimators': 200, 'max_depth': 3, 'min_samples_leaf': 2} (A8 benchmark)	F1	-297.003634	31.737238	31.683702	-280.387	-311.26

Primary model (Ridge F1, $\alpha=0.01$): holdout RMSE = 1.340287, $R^2 = 0.46853$. Relative to OLS F0: $\Delta R^2 = +17.086$ pp, Δ RMSE = +82.63%.

Computational cost comparison: OLS and Ridge training on $n=9$ observations takes under 1 millisecond. Random Forest (100 estimators) runs in approximately 50–200 ms. On the full dataset (hundreds of practices), Ridge will remain computationally trivial while Random Forest may require several minutes for grid search. Computational cost does not differentiate the models at thesis scale.

Methodological caveat: all numerical results in this assignment are computed on a demonstration dataset of 12 observations and 4 practices. Negative R^2 values are expected

when the test partition contains a single practice whose residuals are large relative to the test-set variance. These results validate the pipeline and methodology; substantive conclusions require the full Public Health Scotland analytical table.

F. Interpretation in Relation to Research Question

Research question (Assignment 1): Does including the demographic structure of GP practice populations allow better explanation of variation in hospital service use compared with models using only basic practice characteristics?

Hypothesis: Including `share_age_65_plus` and demographic controls will improve R^2 by ≥ 5 pp and reduce RMSE by $\geq 5\%$ relative to OLS F0.

Assessment on the demonstration dataset: The holdout test set contains a single practice and three observations, making the numeric success criteria unreliable as a substantive test. However, the cross-validation evidence is informative: Ridge F1 achieved a CV RMSE of approximately 8.3, compared to an OLS F0 CV RMSE of 43.9 — a reduction of approximately 81%. This strongly supports the hypothesis that demographic variables carry substantial explanatory value beyond practice size and GP availability.

The direction of findings is consistent with Assignment 2 literature. Gray et al. (2017) showed that age becomes the dominant predictor of hospital admissions after 65. Busby et al. (2017) documented that deprivation is a significant practice-level predictor of unplanned admissions for ambulatory care sensitive conditions. The positive coefficient on `share_age_65_plus` (if estimated on a meaningful sample) would directly confirm that practices with older populations generate higher hospital use per 1,000 patients — the central claim of the thesis hypothesis.

The research gap identified in Assignment 2 — the absence of practice-level models in Scotland that transparently compare a baseline and a demographic-extended specification under consistent experimental conditions — is addressed by the methodological design implemented in Assignments 7 and 8. The pipeline is fully reproducible and ready for the full dataset without modification.

The nonlinear benchmark (Random Forest) did not consistently outperform Ridge in cross-validation. This is consistent with Assignment 2 conclusions: the practice-level problem favours interpretable regression over complex ML models when sample sizes are modest and the research priority is explanation rather than prediction accuracy.